

Chapter 12: Exponential and Logarithmic Equations

(Phương trình Mũ và Logarit)

Chapter Overview

Exponential and logarithmic equations appear frequently in real-world applications, from calculating investment growth and radioactive decay to determining the time needed for populations to reach certain levels. In this chapter, we develop systematic techniques for solving these equations and inequalities. We'll learn when to use logarithms to "bring down" exponents, how to apply the properties of logarithms to simplify complex expressions, and how to check our solutions carefully to avoid extraneous roots. These problem-solving skills are essential for modeling and analyzing exponential phenomena in science, economics, and everyday life.

Bilingual Vocabulary (Từ vựng Song ngữ)

Basic Concepts (Khái niệm Cơ bản)

English	IPA	Vietnamese	Example
Exponential equation	/ˌɛkspəˈnɛnʃəl ɪˈkweɪʒən/	Phương trình mũ	$2^x = 8$ is an exponential equation.
Logarithmic equation	/ˌlɒɡəˈrɪðmɪk ɪˈkweɪʒən/	Phương trình logarit	$\log_2 x = 3$ is a logarithmic equation.
Exponential inequality	/ˌɛkspəˈnɛnʃəl ɪnɪˈkwɑːləti/	Bất phương trình mũ	$3^x > 9$ is an exponential inequality.
Logarithmic inequality	/ˌlɒɡəˈrɪðmɪk ɪnɪˈkwɑːləti/	Bất phương trình logarit	$\log_5 x < 2$ is a logarithmic inequality.

Solution Methods (Phương pháp Giải)

English	IPA	Vietnamese	Example
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Same base method	/seɪm beɪs 'meθəd/	Phương pháp đưa về cùng cơ số	Convert $4^x = 8$ to $2^{2x} = 2^3$.
Taking logarithms	/'teɪkɪŋ 'lɒːgərɪðəmz/	Lấy logarit	Take log of both sides of $10^x = 50$.
Substitution	/,sʌbstɪ'tuːʃən/	Phép thế	Let $t = 2^x$ to simplify $4^x - 3 \cdot 2^x + 2 = 0$.
Extraneous solution	/ɪk'streɪniəs sə 'luːʃən/	Nghiệm ngoại lai	A solution that doesn't satisfy the original equation.
Domain restriction	/doʊ'meɪn rɪ 'strɪkʃən/	Điều kiện xác định	For log x , we need $x > 0$.

Key Terms (Thuật ngữ Quan trọng)

English	IPA	Vietnamese	Example
Isolate	/'aɪsələɪt/	Cô lập	Isolate the exponential term before taking logarithms.
Verify	/'verɪfaɪ/	Kiểm tra	Always verify solutions in the original equation.

Approximate	/ə'prɒ:ksɪmeɪt/	Xấp xỉ	Use a calculator to approximate $\log_3 7$.
Exact solution	/ɪg'zækt sə'lu:ʃən/	Nghiệm chính xác	$x = \log_2 5$ is the exact solution.
Numerical solution	/nu:'merɪkəl sə'lu:ʃən/	Nghiệm số	$x \approx 2.322$ is the numerical solution.

12.1 Exponential Equations (Phương trình Mũ)

Basic Exponential Equation

Definition: An exponential equation is an equation in which the variable appears in an exponent.

The basic form is:

$$a^x = b$$

where

$$a > 0$$

,

$$a \neq 1$$

Solution:

- If $b \leq 0$
: No solution (since $a^x > 0$ for all x)

- If
 $b > 0$
 : Unique solution
 $x = \log_a b$

Method 1: Same Base Method

When both sides can be expressed as powers of the same base, we can equate the exponents.

Same Base Property: If

$$a > 0$$

and

$$a \neq 1$$

, then:

$$a^u = a^v \Leftrightarrow u = v$$

Strategy:

1. Express both sides as powers of the same base
2. Set the exponents equal
3. Solve the resulting equation

Method 2: Taking Logarithms

When the same base method is not convenient, take logarithms of both sides.

Strategy:

1. Isolate the exponential expression
 2. Take logarithm (common or natural) of both sides
 3. Use the power rule:
 $\log(a^x) = x \log a$
 4. Solve for the variable
-

12.2 Logarithmic Equations (Phương trình Logarit)

Basic Logarithmic Equation

Definition: A logarithmic equation is an equation involving logarithms of variable expressions.

The basic form is:

$$\log_a x = b$$

where

$$a > 0$$

,

$$a \neq 1$$

Solution:

$$x = a^b$$

(unique solution)

Method: Same Base Method

Same Base Property: If

$$u, v > 0$$

and

$$a > 0$$

,

$$a \neq 1$$

, then:

$$\log_a u = \log_a v \Leftrightarrow u = v$$

Strategy:

1. **Check domain:** Identify restrictions (arguments must be positive)
2. **Combine logarithms:** Use properties to get single logarithm on each side
3. **Equate arguments:** If bases are equal, set arguments equal
4. **Solve and verify:** Check solutions against domain restrictions

⚠ Critical: Always check that solutions satisfy the domain restrictions. Extraneous solutions often arise in logarithmic equations.

12.3 Exponential Inequalities (Bất phương trình Mũ)

Basic Exponential Inequality

The basic form is:

$$a^x > b$$

(or

$$\geq$$

,

$$<$$

,

$$\leq$$

) where

$$a > 0$$

,

$$a \neq 1$$

Solution depends on the base:

Case 1: When

$$a > 1$$

- $a^x > b$

:

- If

$$b \leq 0$$

: Solution is all real numbers (

$$x \in \mathbb{R}$$

)

- If

$$b > 0$$

: Solution is

$$x > \log_a b$$

Case 2: When

$$0 < a < 1$$

- $a^x > b$
:
 - If $b \leq 0$
: Solution is all real numbers ($x \in \mathbb{R}$)
 - If $b > 0$
: Solution is $x < \log_a b$

Key principle: The direction of the inequality **reverses** when the base is between 0 and 1.

Same Base Property for Inequalities:

- If $a > 1$
:
 $a^u > a^v \Leftrightarrow u > v$
- If $0 < a < 1$
:
 $a^u > a^v \Leftrightarrow u < v$

12.4 Logarithmic Inequalities (Bất phương trình Logarit)

Basic Logarithmic Inequality

The basic form is:

$$\log_a x > b$$

(or

$$\geq$$

,

$$<$$

,

$$\leq$$

) where

$$a > 0$$

,

$$a \neq 1$$

Solution depends on the base:

Case 1: When

$$a > 1$$

- $\log_a x > b$

: Solution is

$$x > a^b$$

Case 2: When

$$0 < a < 1$$

- $\log_a x > b$

: Solution is

$$0 < x < a^b$$

Key principle: The direction of the inequality **reverses** when the base is between 0 and 1.

Same Base Property for Inequalities:

- If
 $a > 1$
 :
 $\log_a u > \log_a v \Leftrightarrow u > v > 0$

- If
 $0 < a < 1$
 :
 $\log_a u > \log_a v \Leftrightarrow 0 < u < v$

⚠ Critical: Always maintain domain restrictions ($x > 0$) when solving logarithmic inequalities.

Worked Examples (Ví dụ Mẫu)

Example 1: Exponential Equation - Same Base Method

Problem: Solve

$$3^{x+1} = \frac{1}{3^{1-2x}}$$

Solution:

First, rewrite the right side with base 3:

$$\frac{1}{3^{1-2x}} = 3^{-(1-2x)} = 3^{2x-1}$$

The equation becomes:

$$3^{x+1} = 3^{2x-1}$$

Since the bases are equal, we can equate the exponents:

$$x + 1 = 2x - 1$$

Solving for

x

:

$$1 + 1 = 2x - x$$

$$2 = x$$

Answer:

$$x = 2$$

Verification:

$$3^{2+1} = 3^3 = 27$$

and

$$\frac{1}{3^{1-4}} = \frac{1}{3^{-3}} = 3^3 = 27$$

✓

Example 2: Exponential Equation - Taking Logarithms

Problem: Solve

$$10^{x-1} = 2022$$

Solution:

Take the common logarithm of both sides:

$$\log(10^{x-1}) = \log(2022)$$

Using the power rule:

$$(x - 1) \log 10 = \log 2022$$

Since

$$\log 10 = 1$$

:

$$x - 1 = \log 2022$$

Therefore:

$$x = 1 + \log 2022$$

Numerical approximation: Using a calculator,

$$\log 2022 \approx 3.306$$

, so

$$x \approx 4.306$$

Answer:

$$x = 1 + \log 2022 \approx 4.306$$

Example 3: Exponential Equation - Substitution Method

Problem: Solve

$$4^x - 3 \cdot 2^x + 2 = 0$$

Solution:

Notice that

$$4^x = (2^2)^x = (2^x)^2$$

. Let

$$t = 2^x$$

(where

$$t > 0$$

).

The equation becomes:

$$t^2 - 3t + 2 = 0$$

Factoring:

$$(t - 1)(t - 2) = 0$$

So

$$t = 1$$

or

$$t = 2$$

Case 1:

$$t = 1$$

$$2^x = 1 = 2^0$$

$$x = 0$$

Case 2:

$$t = 2$$

$$2^x = 2 = 2^1$$

$$x = 1$$

Answer:

$$x = 0$$

or

$$x = 1$$

Verification:

• For

$$x = 0$$

:

$$4^0 - 3(2^0) + 2 = 1 - 3 + 2 = 0$$

✓

• For

$$x = 1$$

:

$$4^1 - 3(2^1) + 2 = 4 - 6 + 2 = 0$$

✓

Example 4: Logarithmic Equation - Basic Form

Problem: Solve

$$4 + 3 \log(2x) = 16$$

Solution:

Step 1: Check domain:

$$2x > 0$$

, so

$$x > 0$$

Step 2: Isolate the logarithm:

$$3 \log(2x) = 12$$

$$\log(2x) = 4$$

Step 3: Convert to exponential form:

$$2x = 10^4 = 10,000$$

$$x = 5,000$$

Step 4: Verify domain:

$$5,000 > 0$$

✓

Answer:

$$x = 5,000$$

Example 5: Logarithmic Equation - Same Base Method

Problem: Solve

$$\log_3(x + 1) = \log_3(x^2 - 1)$$

Solution:

Step 1: Check domain:

- $x + 1 > 0 \Rightarrow x > -1$
- $x^2 - 1 > 0 \Rightarrow (x - 1)(x + 1) > 0 \Rightarrow x < -1$
or
 $x > 1$

Combining:

$$x > 1$$

Step 2: Since bases are equal, equate arguments:

$$x + 1 = x^2 - 1$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

So

$$x = 2$$

or

$$x = -1$$

Step 3: Check against domain (

$$x > 1$$

):

- $x = 2$

: satisfies

$$x > 1$$

✓

- $x = -1$

: does not satisfy

$$x > 1$$

X (extraneous)

Answer:

$$x = 2$$

Example 6: Logarithmic Equation - Product Rule

Problem: Solve

$$\log_2(x + 2) + \log_2(x - 1) = 1$$

Solution:

Step 1: Check domain:

- $x + 2 > 0 \Rightarrow x > -2$

- $x - 1 > 0 \Rightarrow x > 1$

Combined domain:

$$x > 1$$

Step 2: Use product rule:

$$\log_2[(x+2)(x-1)] = 1$$

Step 3: Convert to exponential form:

$$(x+2)(x-1) = 2^1 = 2$$

Step 4: Expand and solve:

$$x^2 + x - 2 = 2$$

$$x^2 + x - 4 = 0$$

Using the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1+16}}{2} = \frac{-1 \pm \sqrt{17}}{2}$$

$$x = \frac{-1 + \sqrt{17}}{2} \approx 1.56$$

or

$$x = \frac{-1 - \sqrt{17}}{2} \approx -2.56$$

Step 5: Check domain (

$$x > 1$$

):

$$\cdot x = \frac{-1 + \sqrt{17}}{2} \approx 1.56 > 1$$

✓

$$\cdot x = \frac{-1 - \sqrt{17}}{2} \approx -2.56 < 1$$

X (extraneous)

Answer:

$$x = \frac{-1 + \sqrt{17}}{2}$$

Example 7: Exponential Inequality

Problem: Solve

$$16^x > \frac{1}{8}$$

Solution:

Express both sides with base 2:

$$16^x = (2^4)^x = 2^{4x}$$

$$\frac{1}{8} = \frac{1}{2^3} = 2^{-3}$$

The inequality becomes:

$$2^{4x} > 2^{-3}$$

Since the base

$$2 > 1$$

, the exponential function is increasing, so:

$$4x > -3$$

$$x > -\frac{3}{4}$$

Answer:

$$x > -\frac{3}{4}$$

or

$$x \in \left(-\frac{3}{4}, +\infty\right)$$

Example 8: Logarithmic Inequality

Problem: Solve

$$\log_{0.3}(x+1) \leq \log_{0.3}(2x-1)$$

Solution:

Step 1: Check domain:

$$\cdot \quad x+1 > 0 \Rightarrow x > -1$$

$$\cdot \quad 2x-1 > 0 \Rightarrow x > \frac{1}{2}$$

Combined domain:

$$x > \frac{1}{2}$$

Step 2: Since the base

$$0.3 < 1$$

, the logarithmic function is decreasing, so the inequality **reverses**:

$$x+1 \geq 2x-1$$

$$2 \geq x$$

$$x \leq 2$$

Step 3: Combine with domain:

$$\frac{1}{2} < x \leq 2$$

Answer:

$$x \in (\frac{1}{2}, 2]$$

Applications (Ứng dụng Thực tế)

Application 1: Investment Growth

Problem: You invest \$500 million in a bank account with 7.5% annual interest, compounded annually. How long will it take for your investment to grow to at least \$800 million?

Solution:

Using the compound interest formula:

$$A = P(1 + r)^n$$

where

$$P = 500$$

,

$$r = 0.075$$

,

$$A = 800$$

$$500(1.075)^n = 800$$

$$(1.075)^n = \frac{800}{500} = 1.6$$

Taking logarithm of both sides:

$$n \log(1.075) = \log(1.6)$$

$$n = \frac{\log(1.6)}{\log(1.075)}$$

Using a calculator:

$$n = \frac{0.2041}{0.0314} \approx 6.50$$

Since we can only complete full years, we need

$$n = 7$$

years.

Answer: It will take at least **7 years** for the investment to reach \$800 million.

Application 2: Bacterial Growth

Problem: A bacterial culture starts with 500 bacteria and grows at a rate of 40% per hour. The population after

t

hours is given by:

$$N(t) = 500e^{0.4t}$$

After how many hours will the population exceed 80,000 bacteria?

Solution:

We need to solve:

$$500e^{0.4t} > 80,000$$

Dividing by 500:

$$e^{0.4t} > 160$$

Taking natural logarithm:

$$0.4t > \ln(160)$$

$$t > \frac{\ln(160)}{0.4}$$

Using a calculator:

$$t > \frac{5.075}{0.4} \approx 12.69$$

Answer: After approximately **12.7 hours** (or about 12 hours 42 minutes), the bacterial population will exceed 80,000.

Application 3: Radioactive Decay

Problem: A radioactive substance decays according to the formula:

$$m(t) = 13e^{-0.015t}$$

where

$$m(t)$$

is the mass (in kg) remaining after

$$t$$

days.

a) What is the initial mass?

b) After how many days will the mass be reduced to 5 kg?

Solution:

a) At

$$t = 0$$

:

$$m(0) = 13e^0 = 13 \times 1 = 13$$

kg

b) We need to solve:

$$13e^{-0.015t} = 5$$

$$e^{-0.015t} = \frac{5}{13}$$

Taking natural logarithm:

$$-0.015t = \ln\left(\frac{5}{13}\right)$$

$$t = \frac{\ln(5/13)}{-0.015}$$

Using a calculator:

$$t = \frac{-0.9555}{-0.015} \approx 63.7$$

Answer:

a) The initial mass is **13 kg**.

b) After approximately **63.7 days**, the mass will be reduced to 5 kg.

Practice Exercises (Bài tập Luyện tập)

Exercise 1: Solve the exponential equations:

a)

$$3^{x-1} = 27$$

b)

$$2^{3x-1} = 27$$

c)

$$5^x = 32^{x-1}$$

Exercise 2: Solve the logarithmic equations:

a)

$$\log(x+1) = 2$$

b)

$$\ln x + \ln(x-1) = \ln(4x)$$

c)

$$\log_3(x^2 - 3x + 2) = \log_3(2x - 4)$$

Exercise 3: Solve the exponential inequalities:

a)

$$0.1^{2x-1} \leq 0.1^{2-x}$$

b)

$$3 \cdot 2^{x+1} \leq 1$$

Exercise 4: Solve the logarithmic inequalities:

a)

$$\log_3(x+7) \geq -1$$

b)

$$\log_{0.5}(x+7) \geq \log_{0.5}(2x-1)$$

Review Exercises (Bài tập Ôn tập)

Exercise 1: Mixed exponential equations

Solve:

a)

$$4^x - 5 \cdot 2^x + 4 = 0$$

b)

$$9^x - 4 \cdot 3^x + 3 = 0$$

c)

$$e^{2x} - 3e^x + 2 = 0$$

Exercise 2: Complex logarithmic equations

Solve:

a)

$$\log_2 x + \log_2(x - 6) = 4$$

b)

$$\log(x + 3) + \log(x - 2) = 1$$

c)

$$\ln(x + 1) - \ln(x - 1) = \ln 3$$

Exercise 3: Real-world applications

a) A car purchased for \$780 million depreciates according to

$$V(t) = 780(0.905)^t$$

, where

t

is in years. After how many years will the car's value drop below \$300 million?

b) The temperature

T

(in °C) of an object decreases according to

$$T = 25 + 70e^{-0.5t}$$

, where

t

is in minutes. After how long will the temperature reach 30°C?

Exercise 4: Challenging problems

a) Solve:

$$2^x + 2^{-x} = 3$$

b) Solve:

$$\log_2 x + \log_4 x + \log_8 x = 11$$

c) Solve the system:

$$\begin{cases} 2^x \cdot 3^y = 12 \\ 2^{x+1} \cdot 3^{y-1} = 4 \end{cases}$$

Chapter Summary (Tóm tắt Chương)

Key Concepts

1. Exponential Equations

- Basic form:

$$a^x = b$$

- Solution:

$$x = \log_a b$$

(if

$$b > 0$$

)

- Methods: Same base, taking logarithms, substitution

2. Logarithmic Equations

- Basic form:

$$\log_a x = b$$

- Solution:

$$x = a^b$$

- Always check domain restrictions
- Watch for extraneous solutions

3. Exponential Inequalities

- When
 $a > 1$
 :
 $a^u > a^v \Leftrightarrow u > v$

- When
 $0 < a < 1$
 :
 $a^u > a^v \Leftrightarrow u < v$

- Direction reverses for base < 1

4. Logarithmic Inequalities

- When
 $a > 1$
 :
 $\log_a u > \log_a v \Leftrightarrow u > v > 0$

- When
 $0 < a < 1$
 :
 $\log_a u > \log_a v \Leftrightarrow 0 < u < v$

- Always maintain
 $x > 0$

Solution Strategies

Equation Type	Strategy
$a^x = a^y$	Equate exponents: $x = y$
$a^x = b$	Take logarithm: $x = \log_a b$

$\log_a x = \log_a y$	Equate arguments: $x = y$ (check $x, y > 0$)
$\log_a x = b$	Convert to exponential: $x = a^b$
Quadratic in a^x	Substitute $t = a^x$, solve for t , then for x

Critical Reminders

- ✓ **Always check domain restrictions** for logarithmic equations and inequalities
- ✓ **Verify solutions** by substituting back into the original equation
- ✓ **Watch for extraneous solutions**, especially in logarithmic equations
- ✓ **Remember inequality direction reverses** when base is between 0 and 1
- ✓ **Use exact form** (e.g.,
 $x = \log_2 5$
) unless approximation is requested

Common Pitfalls

- ✗ Forgetting domain restrictions:
 $\log x$
requires
 $x > 0$
- ✗ Not checking for extraneous solutions
- ✗ Reversing inequality incorrectly based on base value
- ✗ Applying logarithm properties incorrectly:
 $\log(a + b) \neq \log a + \log b$
- ✗ Confusing
 $\log_a b$

with
 $\frac{\log b}{\log a}$

(the latter is correct for change of base)

 **Looking Ahead:** In the next chapter, we will explore **derivative rules and techniques**, learning how to find rates of change for polynomial, exponential, logarithmic, and trigonometric functions. These differentiation techniques are fundamental tools in calculus for optimization, motion analysis, and understanding function behavior.